

Bacterial STIs:

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- Chlamydia:** The most common bacterial STI, among teens & young adults
LGV
- Gonorrhoea:** Can affect anyone who is sexually active, with higher rates among young adults.
- Syphilis:** Has distinct stages with varying symptoms, and can have long-term consequences if untreated.
- Granuloma inguinale:** Caused by *Klebsiella granulomatis*.
- Chancroid:** Caused by *Haemophilus ducreyi*.
- Mycoplasma genitalium* & *Ureaplasma urealyticum*:** They cause STIs.

Syphilis

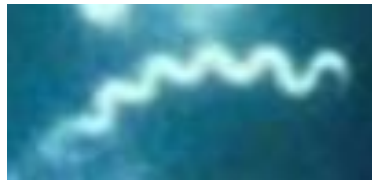
Sexually transmitted infection (STI)

Caused by the bacterium *Treponema pallidum*

(Spirochaete = spiral-shaped) Look at the picture below

It is gram negative but does not stain properly.

Does not grow in bacterial media



Transmission

1. Sexual contact (normal or abnormal): This is the main.
2. Blood transfusion
3. Skin piercing instruments contaminated with infected blood
4. Contact with syphilitic ulcer or rash
5. Congenital (from mother to her foetus in uterus or during delivery)

Stages of Syphilis

1. Primary
 2. Secondary
 3. Latent: infectious
 - early
 - late
 4. Tertiary: non-infectious
- 

Primary Syphilis (highly infectious)

Appears 2 to 6 weeks after infection & remains for 2 to 6 weeks
Characterized by small round firm ulcer(s) 1 to 2 cm known as hard chancre (at the site of infection)

The chancre is neither painful nor tender & of clean base.

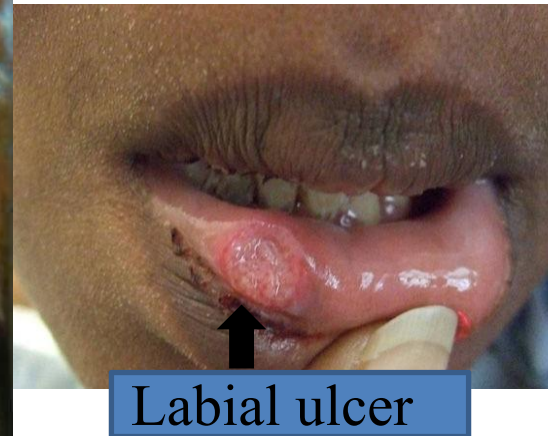
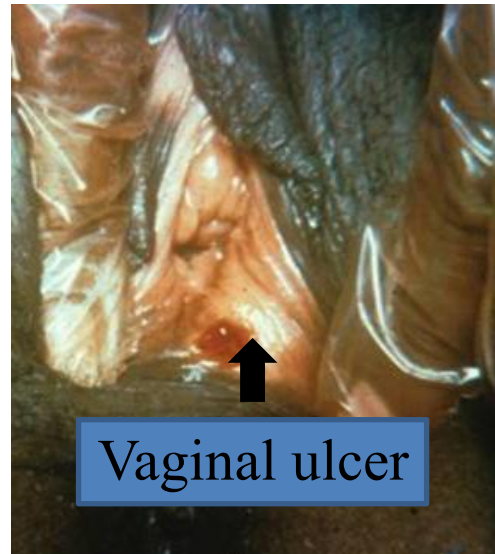
It disappears spontaneously (without treatment)

Occasionally the only manifestation maybe enlarged local lymph nodes.

There maybe multiple lesions when co-infected with HIV

The chancre appears for:

- women in the vagina or cervix
- men in the penis
- in oral sex in the mouth
- in anal sex in the rectum



Secondary Syphilis (highly infectious)

It appears 4 to 10 weeks after the primary stage

Characterized by skin rash & may be sore throat

The rash is non-itchy usually on palms, soles & can be anywhere in the body (look at the pictures).

There maybe ulcers in the mouth (snail-track ulcers look at the picture on) or vagina

This stage disappears without treatment after 3 to 6 weeks.



Snail-track ulcers
in the tongue



Skin rash in the back,



palms



sole

Latent Syphilis (infectious)

This stage is symptomless but serologically proven.

It lasts for years

Divided into:

1. early (a year or less after infection)
2. late (more than a year after infection)

Tertiary Syphilis

It appears 3 to 15 years or even decades after the initial infection

Characterized by gumma formation (granuloma) & neurovascular manifestations.

It has 3 forms:

1. Gummatous (benign syphilis): 1 to 46 years after

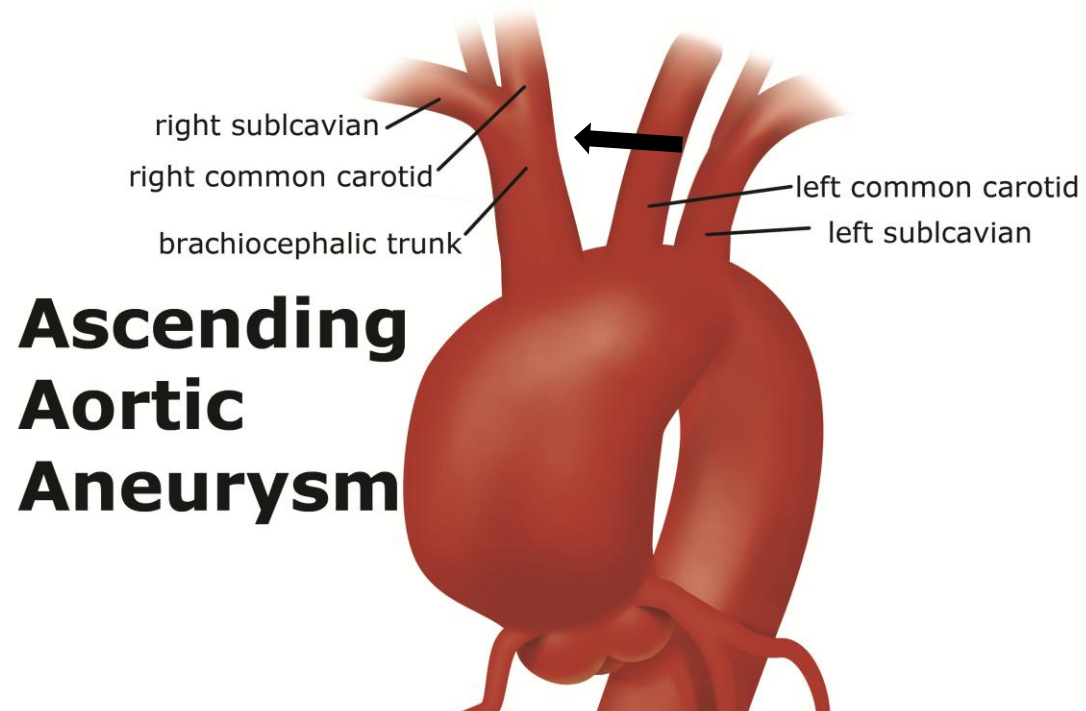
Characterized by gumma formation (granulomas) which are soft tumor-like balls of inflammation: can occur anywhere in the body but commonly in skin, bone & liver.

Gumma



Tertiary Syphilis

2. Cardiovascular: Occurs 20 to 30 years after infection, manifested as syphilitic aortitis that may lead to aortic aneurysm.



Tertiary Syphilis

3. Neurosyphilis:

It can occur at any stage of the disease.

If it occurs early it maybe symptomless or manifests as meningitis.

If it occurs late it manifests as:

- a. Meningovascular syphilis: Inflammation of the vasculature supplying CNS resulting in ischaemia
- b. Tabes dorsalis: Muscle weakness & abnormal sensation
- c. General paresis: Neuropsychiatric disorder

Congenital Syphilis

The patient is the foetus in the uterus

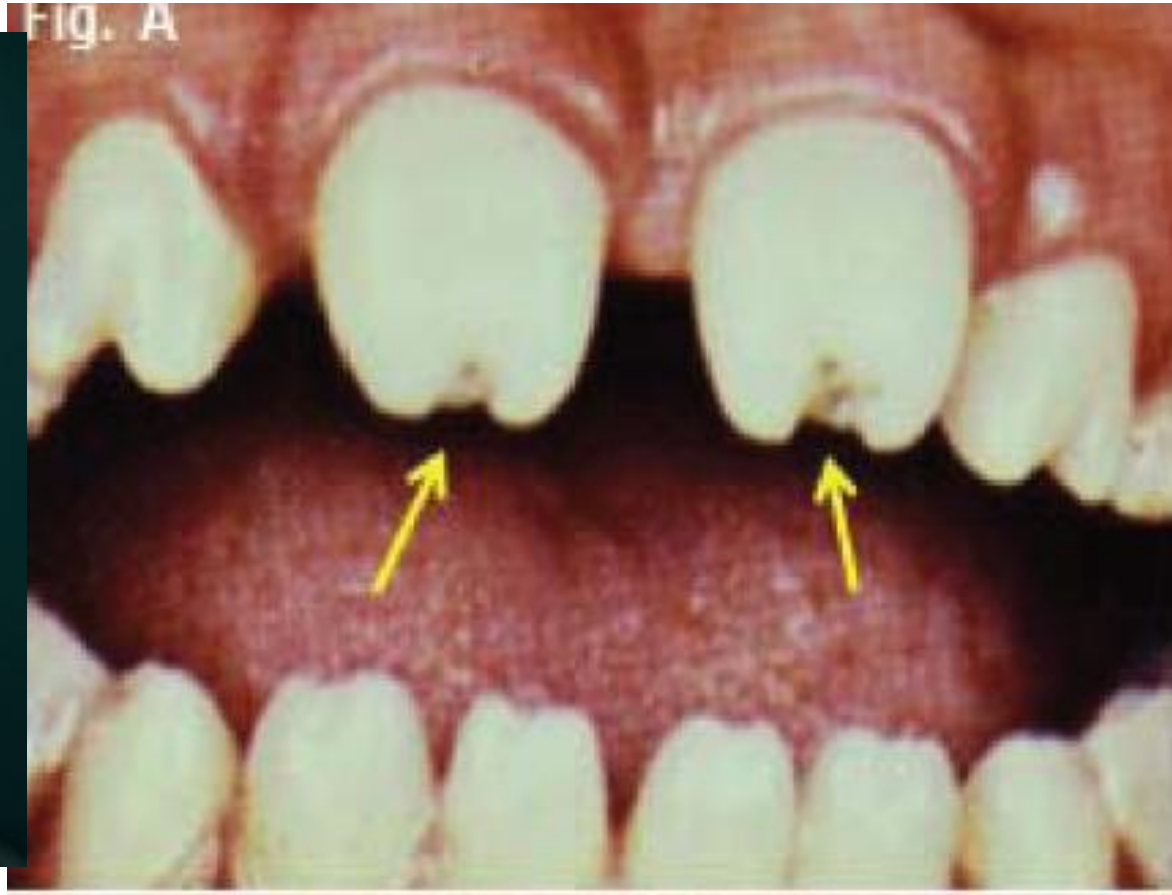
The infection is by transplacental transmission

It may lead to:

1. Miscarriage: loss of early pregnancy
2. Premature birth: Born before 37th week of gestation.
3. Stillbirth: Born full-time but dead.
4. Live birth with congenital syphilis

Hutchinson's triad of congenital syphilis

- i. Hutchinson's teeth: (centrally notched, widely-spaced peg-shaped upper central incisors)
- ii. Deafness
- iii. Interstitial keratitis



Hutchinson's teeth

Diagnosis

Clinical

NB: *T. pallidum* does not grow in culture media

-Demonstration of the bacteria from serous fluid from chancres by:

Dark field microscopy

Direct fluorescent antibody

Polymerase chain reaction



Antibody detection:

- Non-treponemal tests : (Non-specific tests):

Venereal Disease Research Laboratory Test (VDRL)

Rapid Plasma Reagin (RPR)

-Treponemal tests: specific tests (confirmatory)

--Treponema pallidum Haemagglutination Assay (TPHA)

--Fluorescent treponemal antibody absorption test (FTA-abs)

Treatment

Early infection:

Benzathine penicillin

Doxycycline

Tetracycline

Ceftriaxone

Late infection: benzyl penicillin

Prevention:

- Avoidance of risky practices
- There is no vaccine

NB: Jarisch-Herxheimer reaction may occur

It appears 1 to 24 hours after initiation of treatment due to released cytokines as a result of the dead bacterial proteins

Jarisch-Herxheimer reaction

